

Natural Language Processing and modern AI

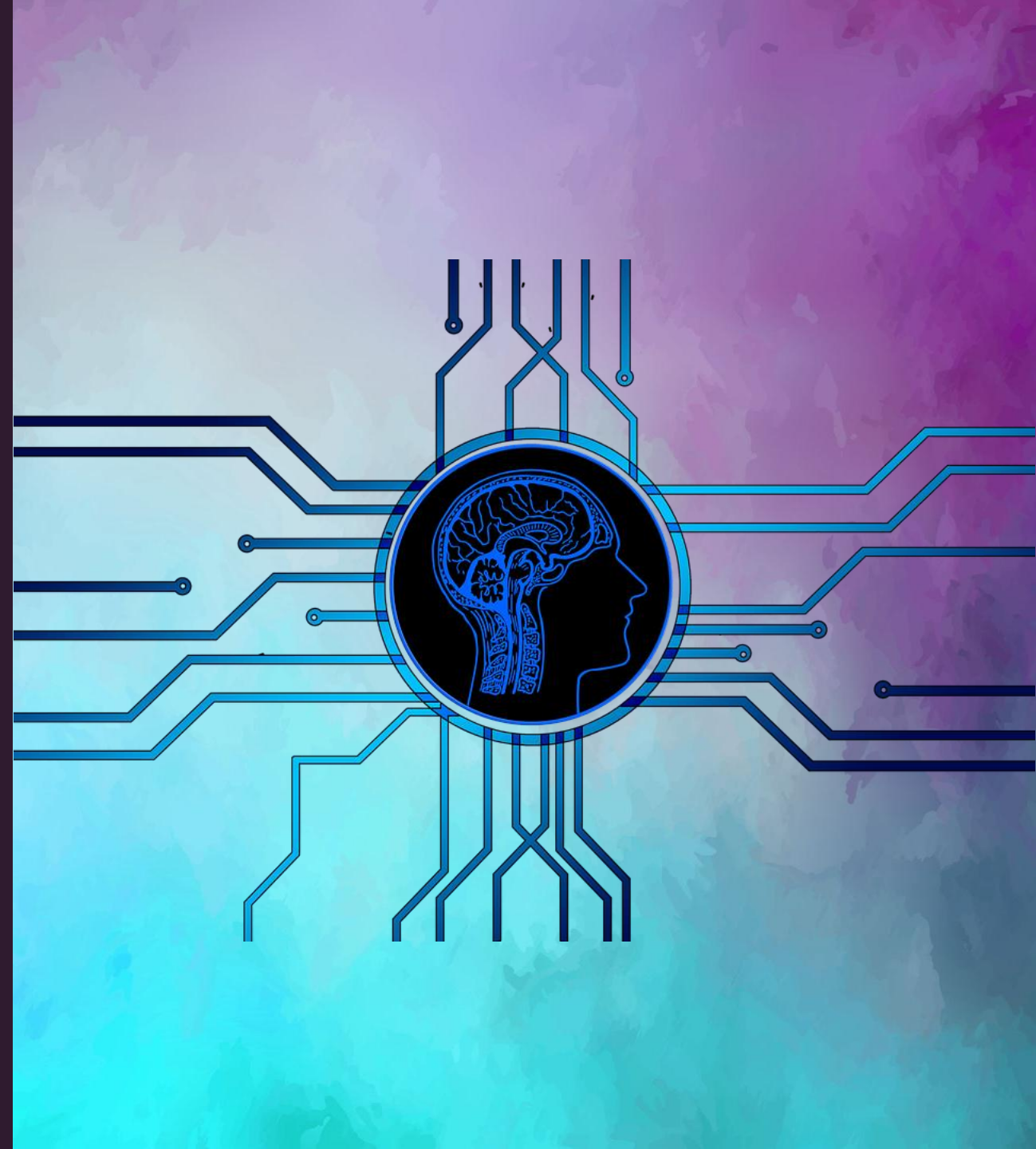
SIT112 | Intro to Data Science & AI
Lecture Week 10

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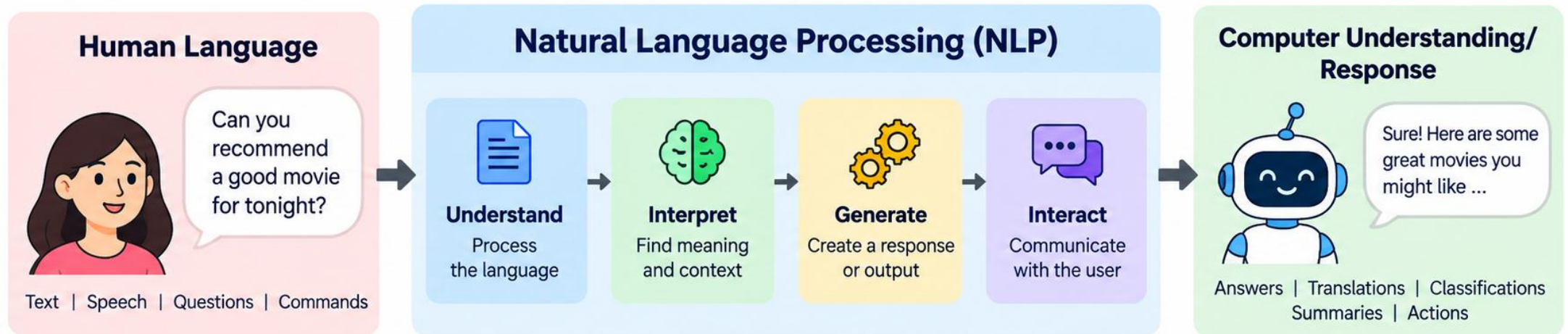
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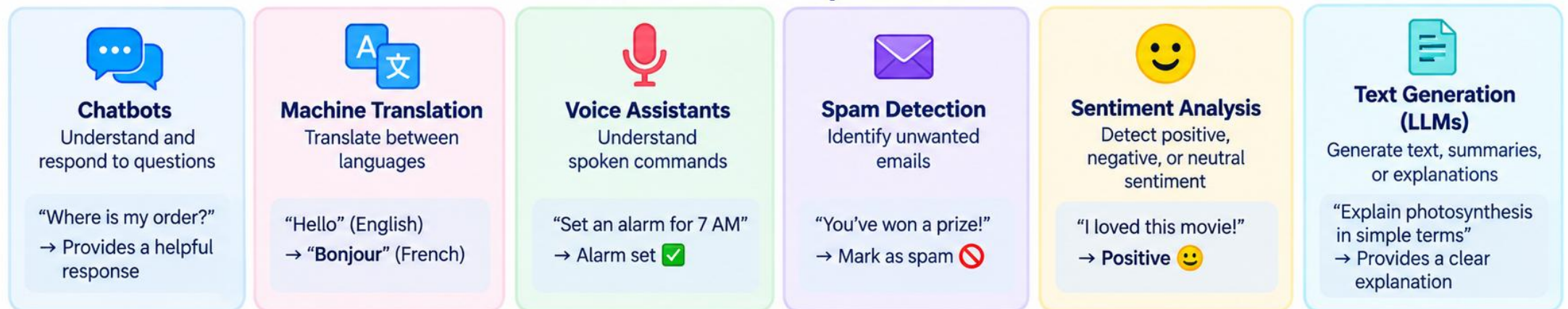
Objectives

- Getting familiar **with Natural language processing**.
- **Understand AI:** Explain what Artificial Intelligence is and distinguish AI from simple rule-based or automated systems.
- **Recognise AI Applications:** Identify how AI is used in everyday life and across areas such as healthcare, education, transport, business and entertainment.

What is Natural Language Processing?



Real-World Examples of NLP

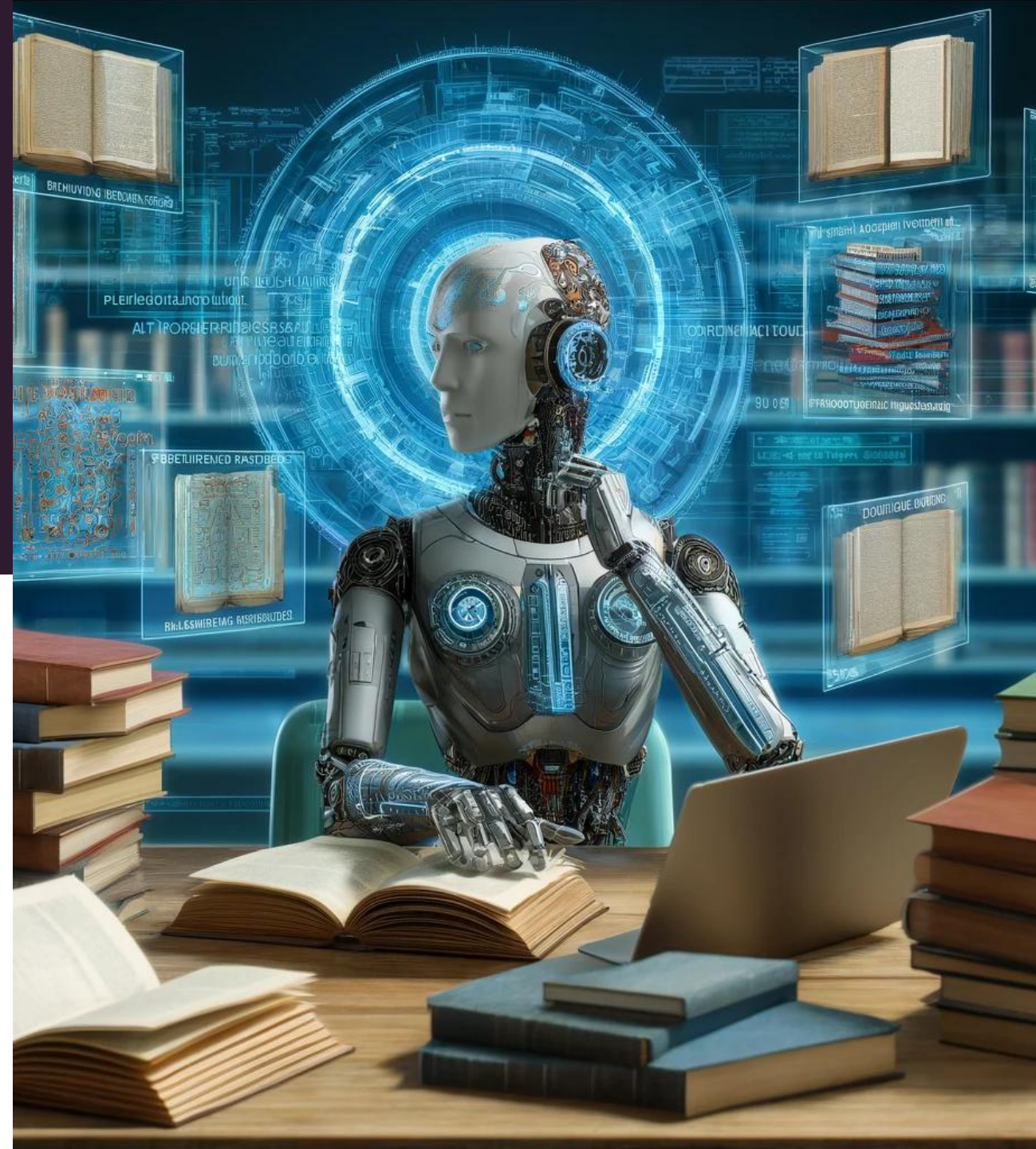


Why NLP?

To Learn from Human Knowledge

Humans **have written down a lot of knowledge** using **natural language**. NLP helps computers learn from **human-written information** and **use it to answer questions**.

Example: An AI system can analyse millions of **Wikipedia pages** to answer, *“What causes climate change?”* **without** someone manually entering every fact into a database.

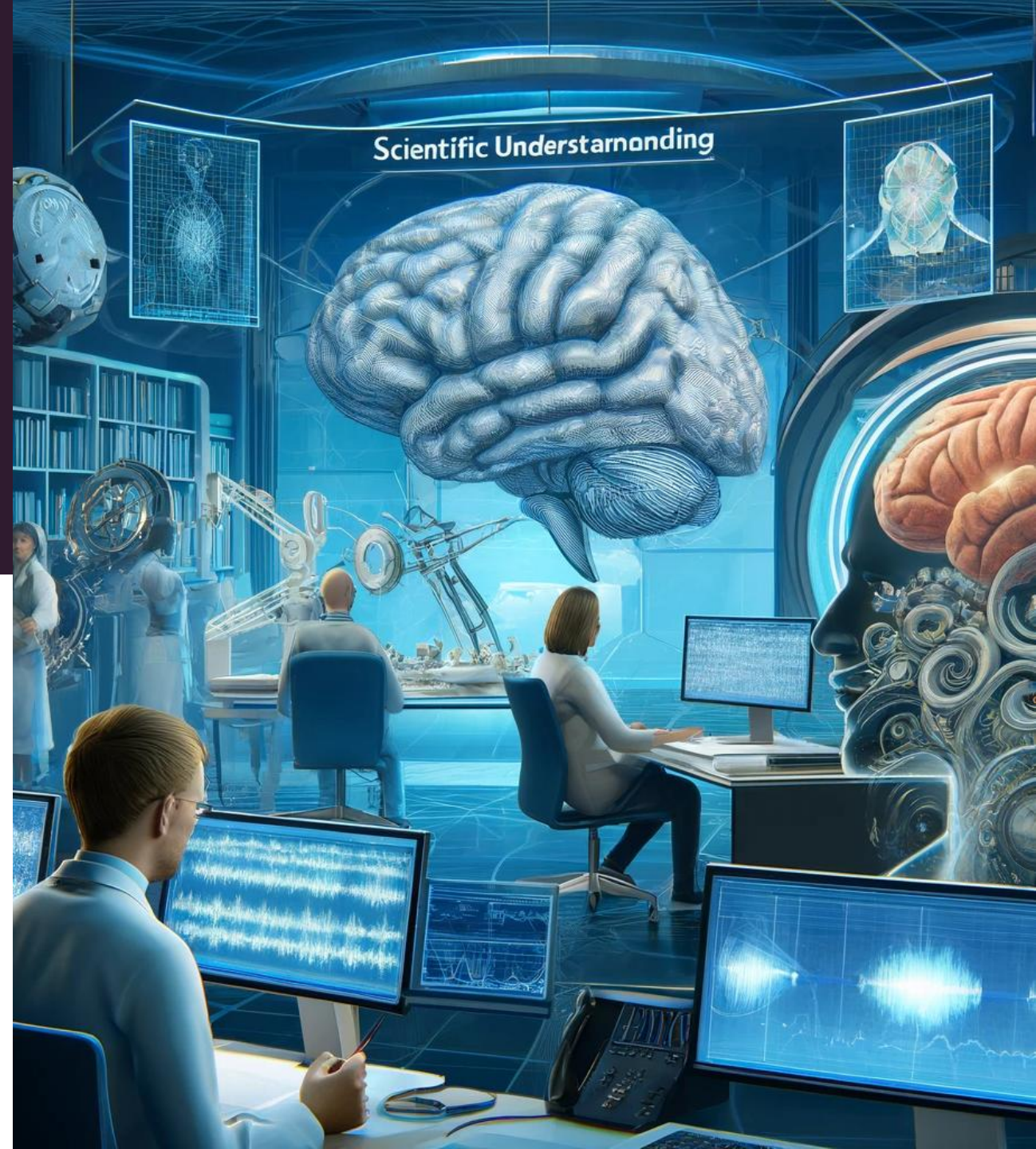


Why NLP?

Convenient interaction:

Voice is useful when typing is difficult; for example, asking a **navigation system** for directions while driving.

Example: Asking **Siri** or **Google Assistant** about the weather.



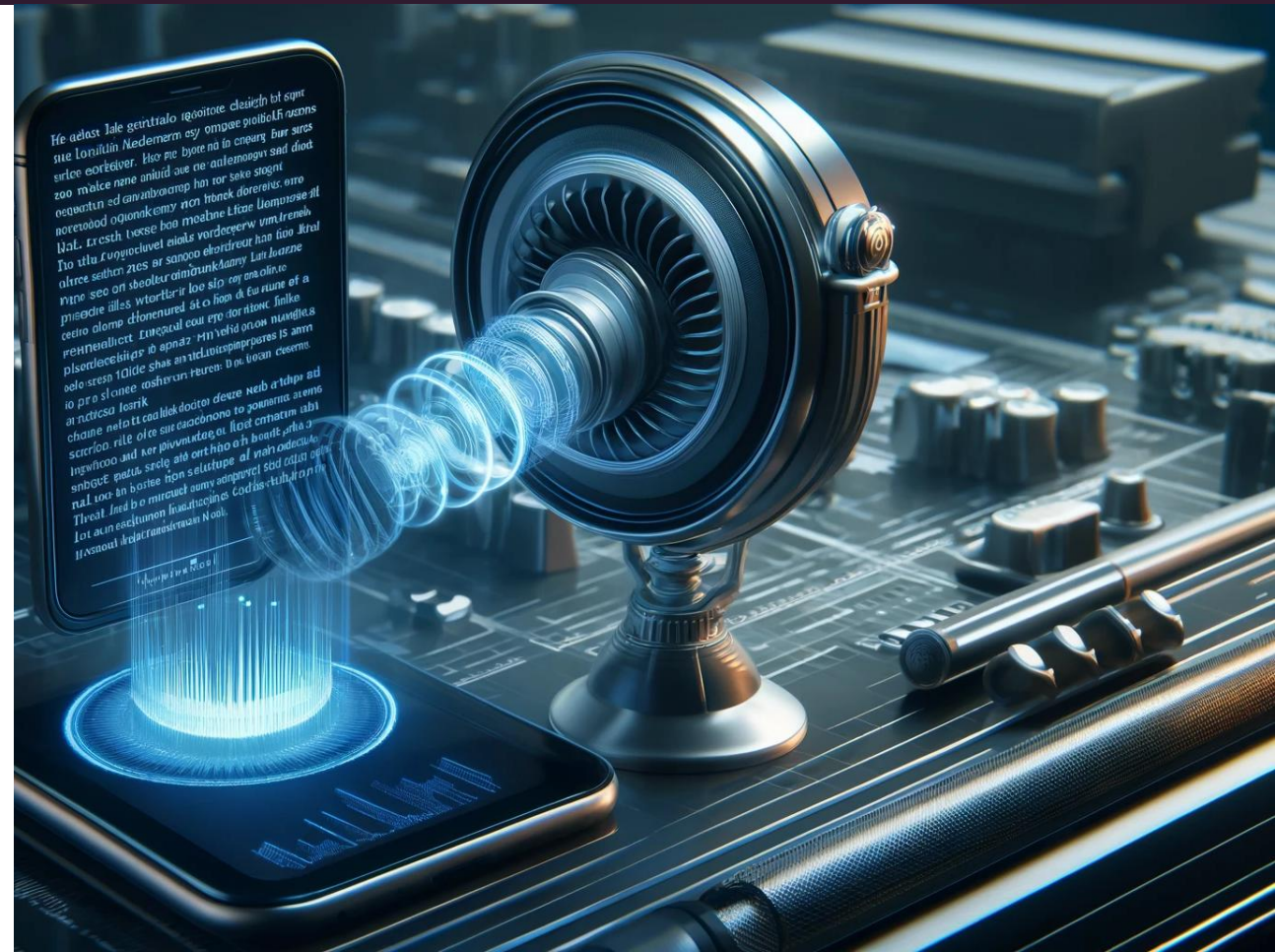
Natural Language Tasks: Speech Recognition

- The task of transforming **spoken sound into text**.
- We can then perform **further tasks** (such as **question answering**) on the resulting text.
- The challenge for a system using speech recognition is to **respond appropriately even when there are errors** in individual words.



Natural Language Tasks: Text-to-Speech

- The challenge is to **pronounce each word correctly** and to make the **flow of each sentence seem natural**, with the **right pauses** and **emphasis**.
- Another area of development is synthesising **different voices**, starting with a choice between a generic **male or female voice**, then allowing for **regional dialects**, and even **imitating celebrity voices**.



Natural Language Tasks: Question Answering

- Question Answering is when the query is a question, such as “Who founded the U.S. Coast Guard?” and the response is not a ranked list of documents but rather an **actual answer**: “Alexander Hamilton.”
- This **differs** from a **traditional search engine**, which normally provides a **ranked list of documents or webpages** that may contain the information.



What Is a Language Models:

It is a system that learns **patterns** in natural language, such as English, from large collections of human-written text.

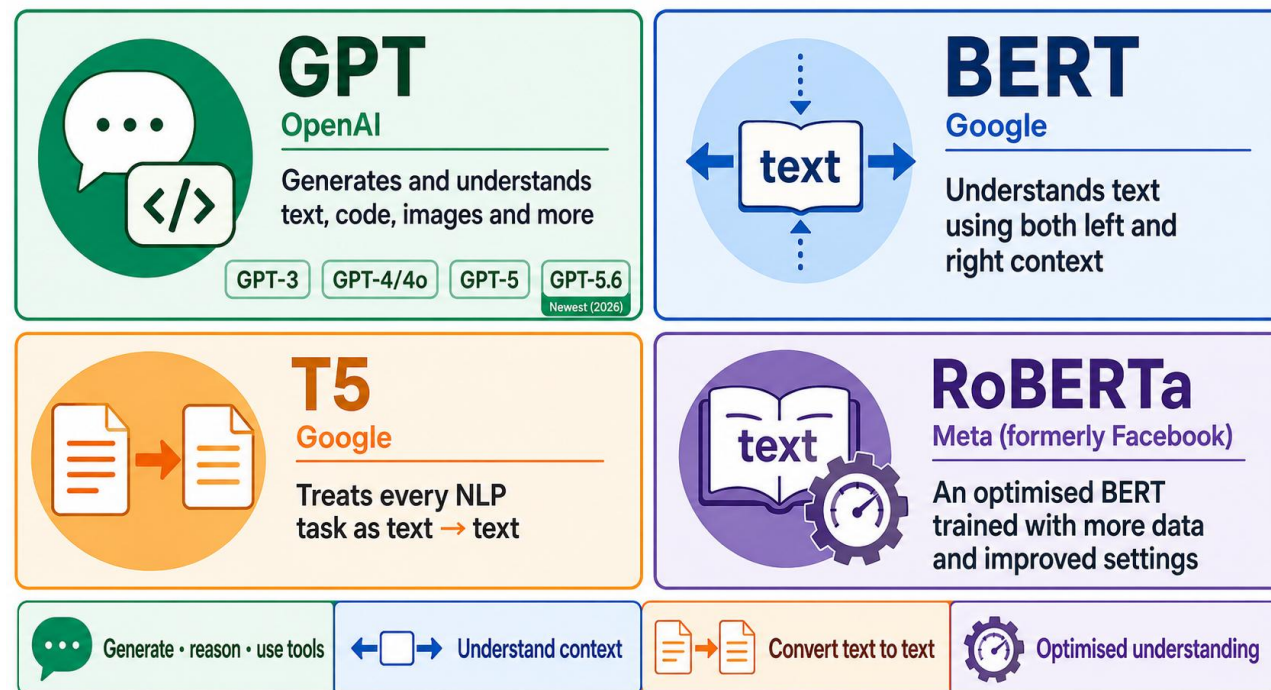
It learns:

- ✓ which words commonly appear together;
- ✓ how sentences are structured;
- ✓ how context changes meaning;
- ✓ which word or phrase is likely to come next.

Examples of Language Models:

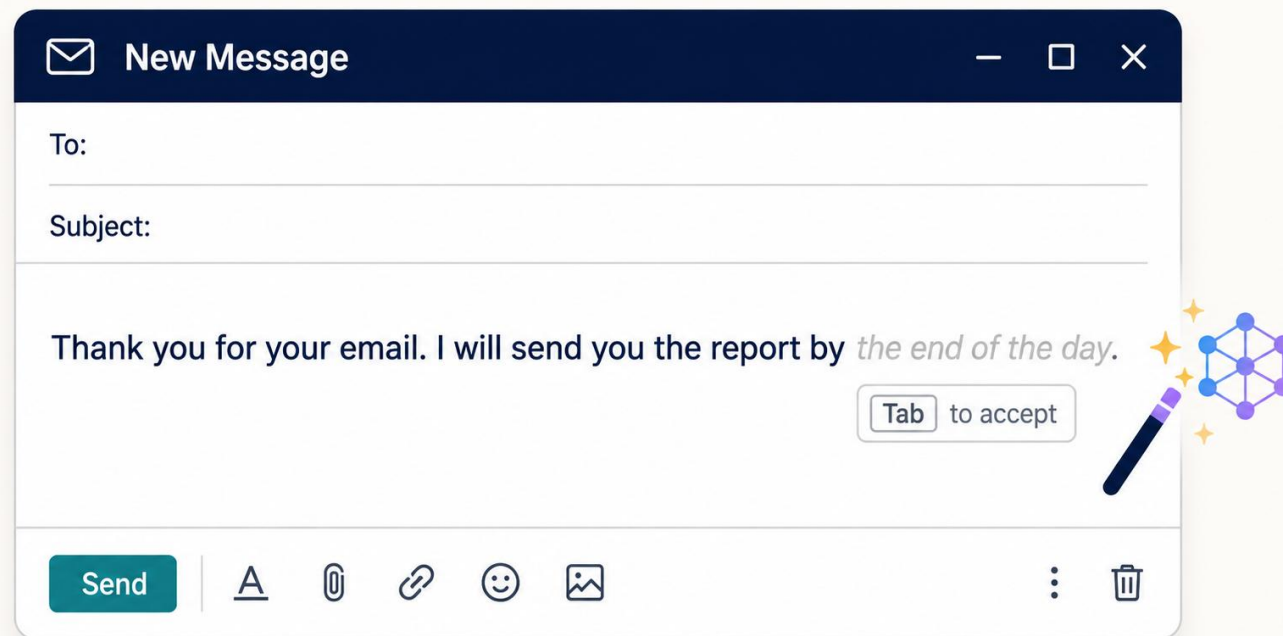
These models support tasks such as **writing**, **translation**, **summarisation**, **classification** and **question answering**.

- **GPT** (Generative Pre-trained Transformer) generates text, code and other content based on a **prompt**.
- **BERT** focus on understanding **context** and information extraction.
- **RoBERTa** is an **optimised** version of BERT trained with changes intended to improve performance.
- **T5** treats many NLP tasks as **text-to-text** problems, meaning it receives text as input and generates text as output.



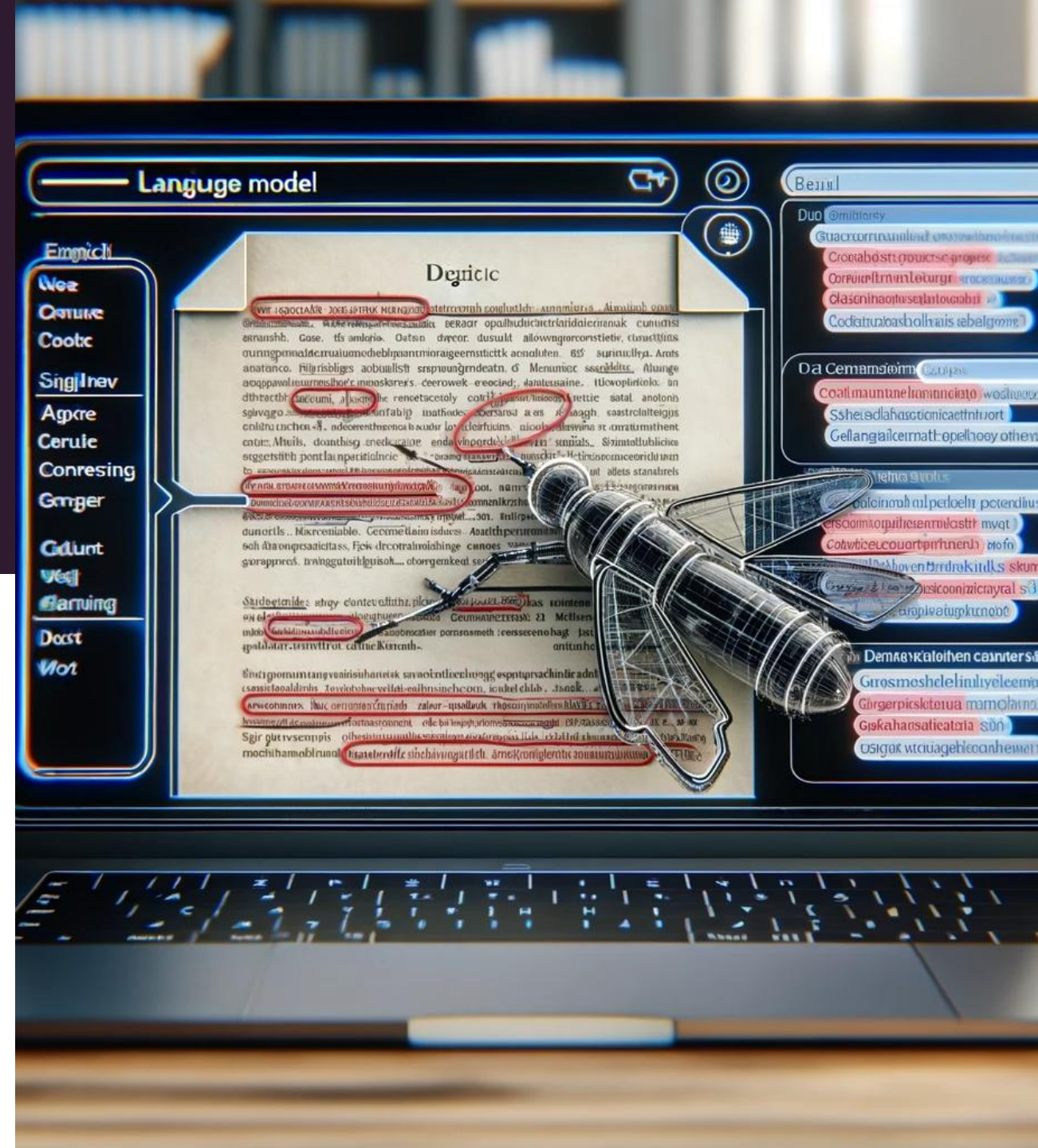
Language Models: Text Prediction and Autocompletion

A language model predicts which words are **most likely** (probability distribution) to **come next** (prediction) and uses these predictions to **suggest completions** for emails or text messages.



Language Models: Spelling and Grammar Check

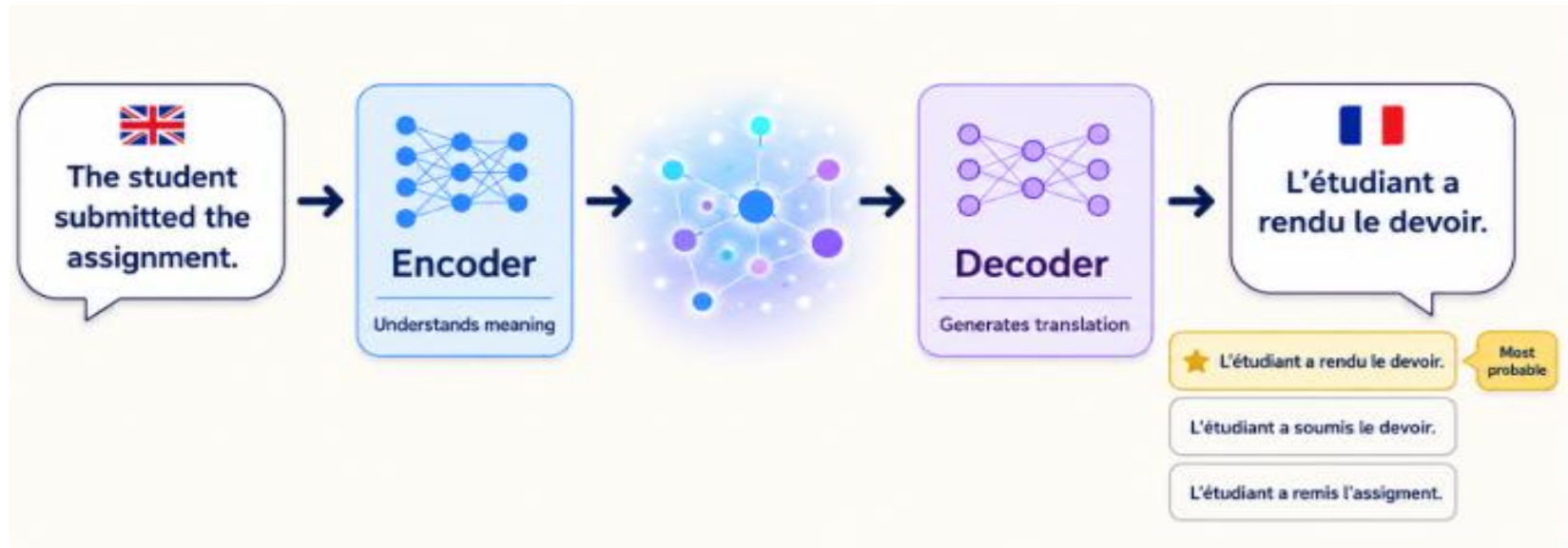
It can identify parts of a sentence that seem **unlikely** or **incorrect** and **suggest changes** that make the text more **natural, grammatically correct** and **probable**.



Language Models: Translation

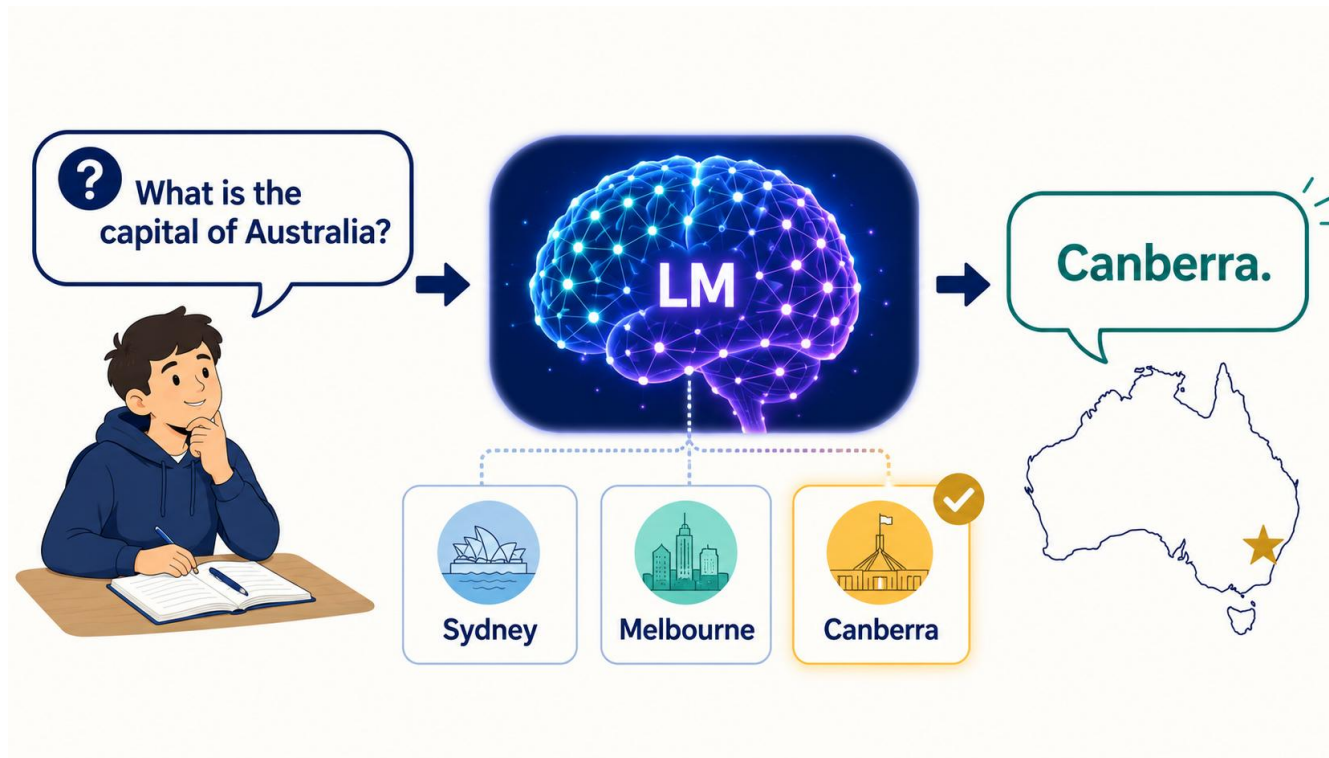
A translation system can use **a pair of models**:

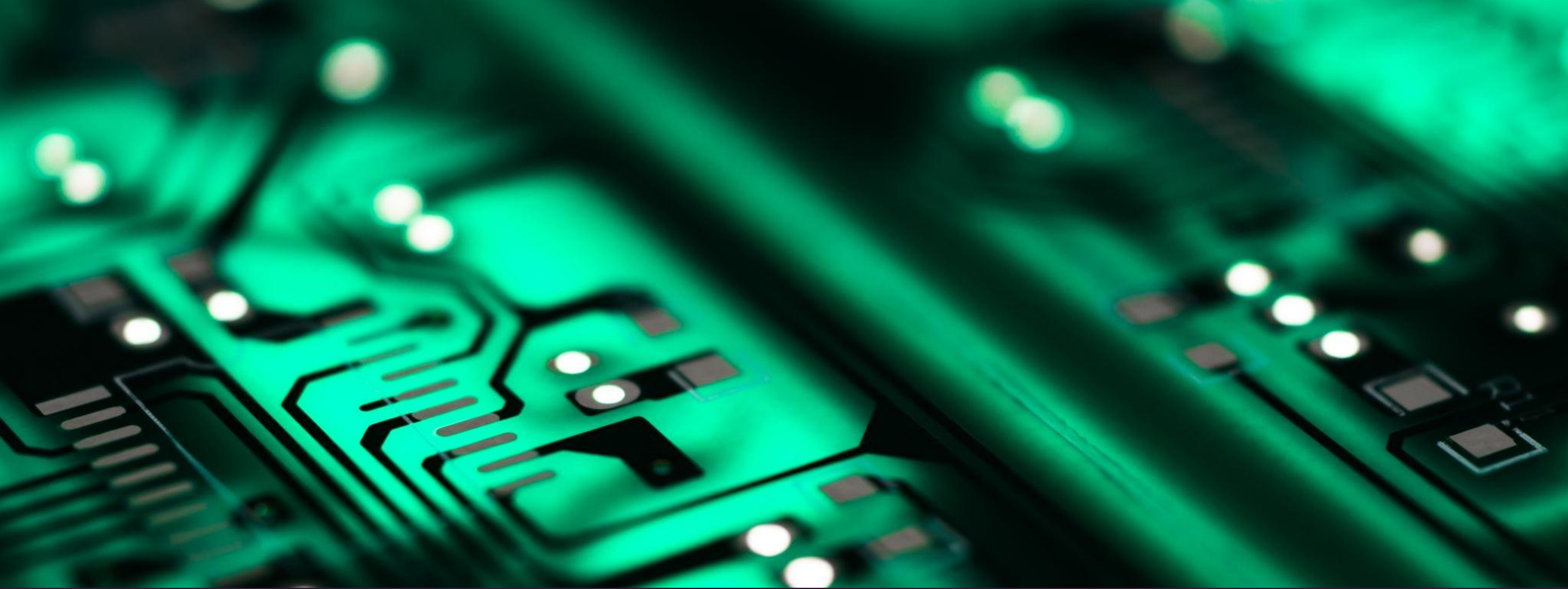
- An **encoder model** reads and understands the sentence in the **original** language.
- A **decoder model** generates the most likely sentence in the **target** language.



Language Models: Question Answering

It can **analyse** a question, **understand** its **context** and **generate** the **most likely** relevant answer.





Case Study: Understanding the Emotional Impacts of Covid-19 News on Twitter (X)

News Avoidance; Reuters Digital News Report, 2022



43%

say there is too
much politics
and COVID-19



36%

say news has
negative effect
on mood



29%

say they are worn
out by amount
of news



29%

say the news is
untrustworthy
or biased



17%

say it leads to
arguments I'd
rather avoid



16%

say there is nothing
I can do with the
information

The Impact of Covid-19 News



- The **media coverage of the pandemic** has triggered a variety of **emotional responses**, which may help or hinder the **effectiveness of public communication**.
- **Psychological distress** resulting from **ongoing media exposure** to the pandemic may impact the **mental health** of people, especially vulnerable groups.



The Goals



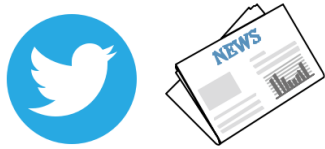
Understanding the emotional response to the media coverage of Covid-19 across the world.



Predicting the impact of Covid-19 news on people's emotions and mental well-being.

The Approach

Covid-19 Information



- Distressing Covid-19 **contents** on social and popular media
- Contagion of **negative sentiments**

Mental Health Impacts



- **Anxiety** levels may be exacerbated by exposure to distressing contents
- People may **abstain** from news/ social media
- Leads to further **isolation** and limited access to information.

Understanding Emotions



- Studied literature that investigated the **impacts of social media** usage and news consumption on **mental health**, amid Covid-19 Pandemic.
- Explored advances in AI and Data Science, e.g., Natural Language Processing for understanding the emotional response to Covid-19.

Researchers with
complementary skills
have worked on this
project.



Psychology



Software Engineering



Data Science

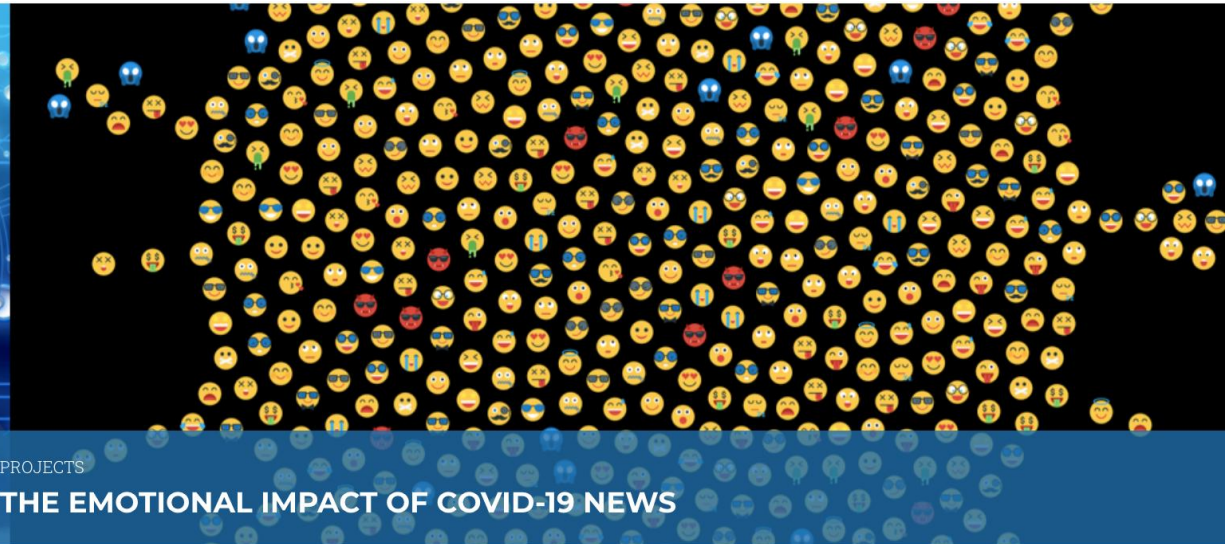


Artificial Intelligence

An Interdisciplinary Project



Investigating the Emotional Response to Covid-19 News on Twitter: A Topic Modeling and Emotion Classification Approach



Overview



To understand people's emotional response to the COVID-19 news, we have investigated the prevalence of basic human emotions in around 19 million user responses to 1.7 million Covid-19 news posts on Twitter from (English-speaking) media across 12 countries from January 2020 to April 2021.

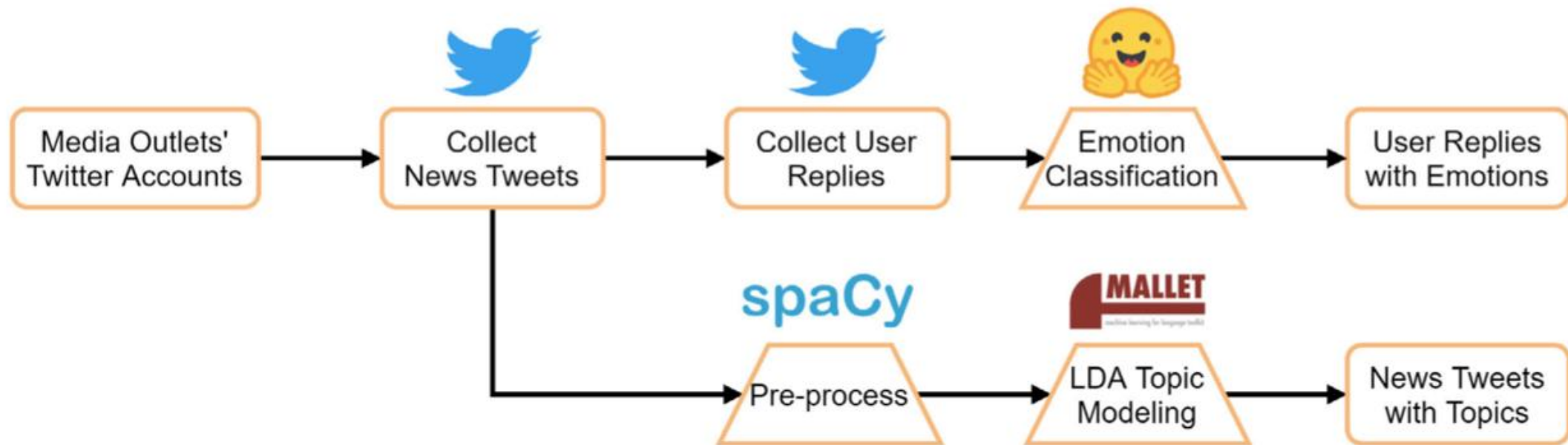


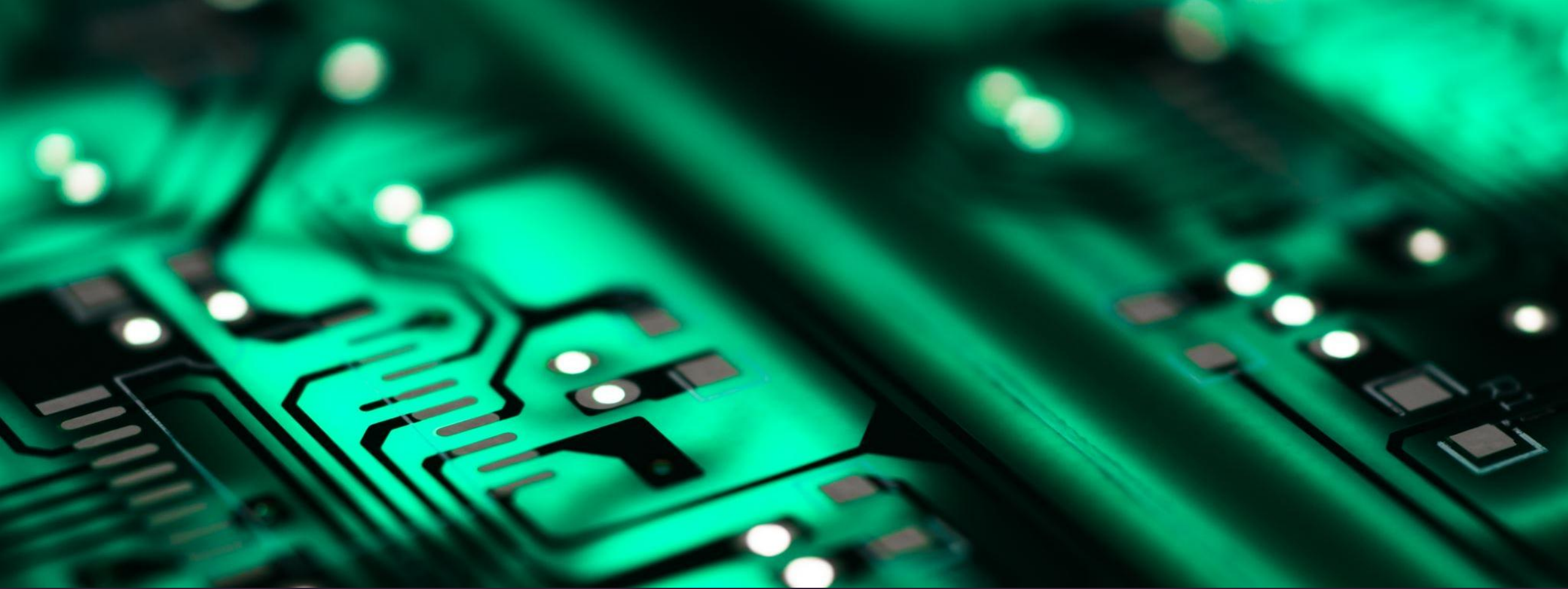
We have used Latent Dirichlet Allocation (LDA) to identify news topics on Twitter.



The Robustly Optimized BERT Pretraining Approach (RoBERTa) model was used to identify emotions in the tweets.

Methodology





Data Collection

The Twitter API

Getting started

How to get access to the Twitter API

Twitter API v2

Tweets

Bookmarks

- [DELETE /2/users/:id/bookmarks/:tweet_id](#)
- [GET /2/users/:id/bookmarks](#)
- [POST /2/users/:id/bookmarks](#)

Filtered stream

- [GET /2/tweets/search/stream](#)
- [GET /2/tweets/search/stream/rules](#)
- [POST /2/tweets/search/stream/rules](#)

Hide replies

- [PUT /2/tweets/:id/hidden](#)

Likes

- [DELETE /2/users/:id/likes/:tweet_id](#)
- [GET /2/tweets/:id/liking_users](#)
- [GET /2/users/:id/liked_tweets](#)
- [POST /2/users/:id/likes](#)

Collecting Twitter Data: News Tweets



- 282 English news accounts were selected based on their popularity and user engagement.
- 6 accounts were removed as they did not publish any news on Covid019.
- A list of keywords was used to collect news tweets and the responses to those tweets from 276 news accounts across 12 countries and 5 continents.

Collecting Twitter Data: News Tweets (Cont.)

- For each news account, we collected the Twitter data for January 1st 2020–April 30th 2021.
- 1,705,830 news tweets were collected containing at least one of the keywords and identified as written in English by Twitter metadata.



Collecting Twitter Data: News Tweets (Cont.)

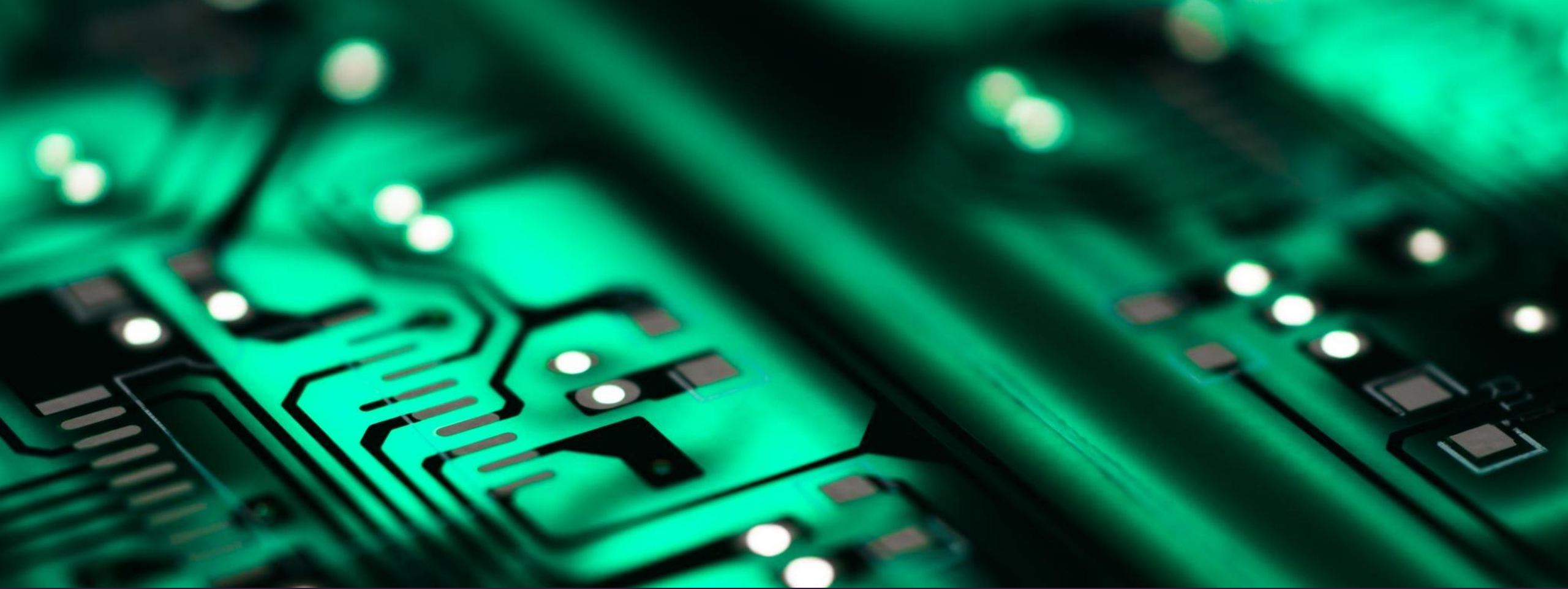


- Some news **tweets** just included a URL **link** to their news articles, **without any text** in the tweet content that we could search for a keyword in.
- In such cases, we **used the URL to fetch the title of the news** and **included only** the tweets with links to news articles whose **headlines** contained a COVID-19-related **keyword**.

Collecting Twitter Data: User Replies



- From each news tweet, we collected the available user replies, **excluding** (i) **replies to replies**, (ii) **deleted** by the author or by the platform, (iii) **not identified as written in English** in Twitter metadata, or (iv) **non-public tweets** from private Twitter accounts.
- This resulted in **18,882,812 user responses**, from **3,038,495 unique Twitter accounts**.



Topic Modeling

Topic Modeling

It is **an NLP technique** used to **automatically** discover the **main themes or topics** within a large collection of documents.

Example:

Suppose we analyse thousands of **student comments**. A topic model may find:

Teaching: lecturer, explanation, class, helpful

Assessment: assignment, deadline, marking, feedback

Technology: software, login, OnTrack, error

How Topic Modelling Works

1. It groups words that frequently appear together and uses them to identify possible topics.
2. The model does not receive these topic names beforehand (unsupervised).
3. A topic-modelling algorithm produces a group of topics, and each topic is represented by a list of related words and a probability for each word.

Example:

Topic 1: assignment (30%), deadline (25%), marking (20%), feedback (15%)

Topic 2: lecture (35%), teacher (25%), explanation (20%), class (10%)

Based on these words, a human might name the topics **Assessment** and **Teaching**.

Topic Modelling: Latent Dirichlet Allocation (LDA).

One popular **algorithm** for topic modelling is Latent Dirichlet Allocation (**LDA**).

LDA **assumes** that **documents are composed of a mixture of topics**, and **each word** in a document is **generated from one of the topics**.

- The **input** to an LDA model is a **text corpus**.
- The **raw** text of a tweet has multiple **issues** when generating a text corpus, such as **different inflections** of the same word (e.g., run, running), **emojis**, **mentions** of other Twitter users and web **links**.
- To address those issues, we **preprocessed** the tweets.

Topic Modeling: Preprocessing

1. **Cleaning the text using Regular Expressions:** We use **Regex** to **keep words and spaces** while **removing numbers and underscore symbols** (regex code: `^[^\\w\\s]|[0-9]|[_]`). For example, suppose the original tweet is:

"COVID_19 cases increased by 25% today!!!"

We may not want numbers, underscores, or punctuation to contribute to our topic analysis. After applying the cleaning rule, we might obtain something like:

"COVID cases increased by today"

The important idea is:

Raw text → Regex → Cleaner text

Why? Because for **topic modelling**, we are primarily interested in **meaningful words** that tell us what the document is about.

Topic Modeling: Preprocessing

2. Converting text into tokens (**tokenization**):

A **token** is a **unit of text** that the model processes. In this example, we can think of tokens mainly as **individual words**. For example:

"people need better healthcare"

becomes:

["people", "need", "better", "healthcare"]

Each item is a token.

So, “**list of tokens**” simply means that instead of keeping the tweet as **one long string**, we **break it into smaller units** that **can be processed**.

Tokenization is like cutting a sentence into pieces that the computer can analyse individually.

Topic Modeling: Preprocessing

3. Removing Twitter usernames

Tweets often contain mentions such as:

"@John I completely agree about climate change."

Here, @John identifies another Twitter account, but the username itself may not tell us much about the topic.

So, we remove tokens beginning with @:

Before:

["@John", "I", "completely", "agree", "about", "climate", "change"]

After:

["I", "completely", "agree", "about", "climate", "change"]

Now **climate** and **change** are useful words for discovering the topic, while @John has been removed as noise.

Topic Modeling: The Number of Topics

LDA is repeatedly run for different numbers of topics: 2, 5, 8, 11, 14, ... up to 40.

Initially, increasing the number of topics substantially improves coherence. But eventually the improvement becomes much smaller, and the curve starts to level off.

But why 20 if the graph gets higher later?

The absolute highest coherence score in the graph appears to occur around 29 topics, not 20. But 20 has been selected as a trade-off between coherence and interpretability/model complexity, because by around 20 topics the curve has largely begun to plateau.

One important characteristic of **LDA** is that we usually have to tell the model **how many topics to find**.

If we choose **too few topics**, the model may **combine very different subjects**. If we choose **too many**, one meaningful topic may be unnecessarily **split** into several **very similar topics**.

What is the coherence score?

Coherence measures how well the important **words within a discovered topic make sense together**.

Imagine LDA produces these two topics:

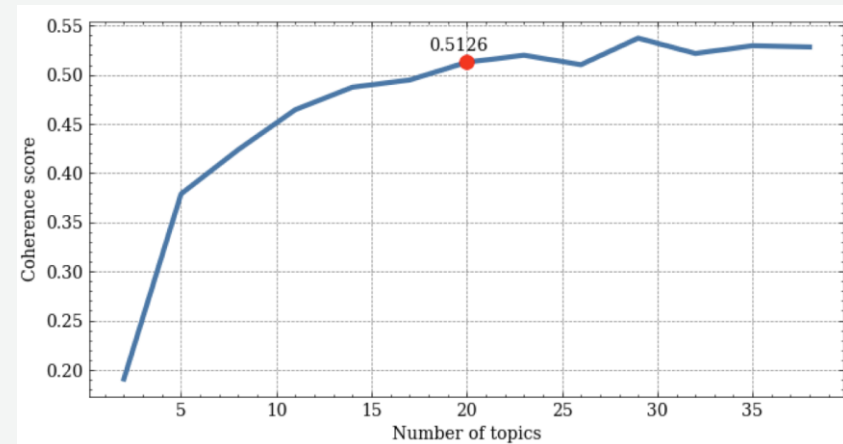
Topic A: hospital, doctor, patient, treatment, disease

Topic B: hospital, election, football, banana, computer

Topic A is much more coherent because its words have a clear relationship. We can easily interpret it as a healthcare topic.

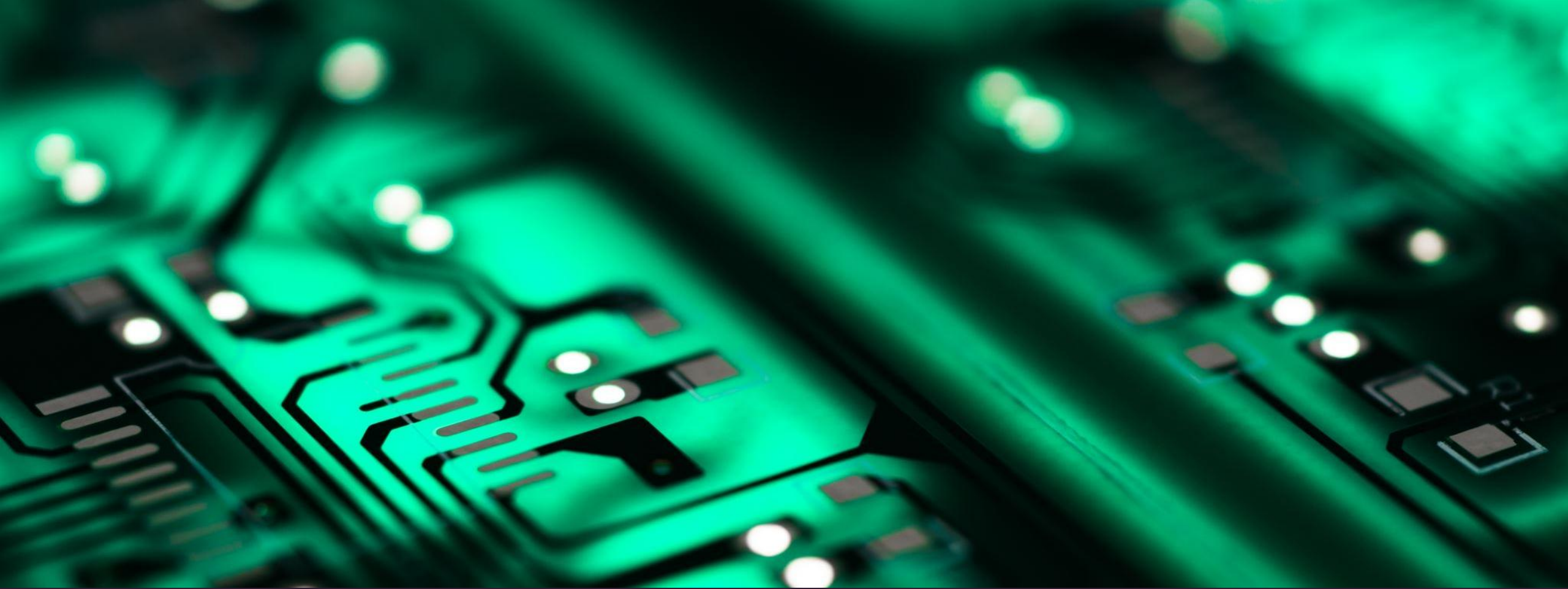
Topic B has unrelated words, so it would have lower coherence. Therefore:

Higher coherence score → generally more meaningful topics



The Identified Topics

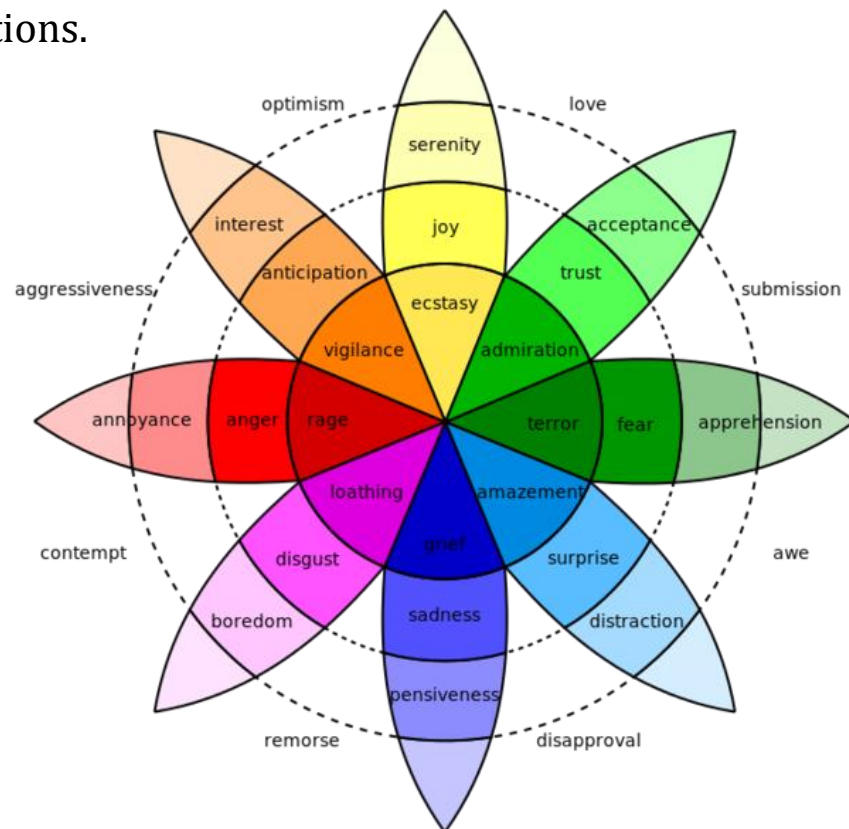
Topic top keywords	Topic Coherence	News tweets prevalence	Themes	Theme Coherence	Subthemes
day, surge, continue, high, country, hit, week, infection, wave, datum	49.34%	6.01%	Cases and deaths	54.99%	(1) Cases and deaths increasing (2) Cases and deaths decreasing
case, report, rise, record, numb, death, infection, daily, confirm, spike	64.86%	5.64%			
case, report, death, total, confirm, hour, recovery, breaking, death_toll, record	72.332%	3.68%			
pandemic, time, face, world, change, year, write, long, life, work	50.76%	5.16%	Economic impact	36.63%	(1) Economic impact over national economies and big companies (2) Economic plans to support jobs and food programs
pandemic, business, economy, impact, year, company, market, demand, employee, support	55.25%	3.84%			
health, vaccination, work, provide, service, community, fight, question, launch, worker	42.98%	2.09%			
amid, pandemic, school, close, student, child, open, fear, concern, shut	47.09%	2.75%	Educational impact	44.92%	None
patient, hospital, die, test_positive, doctor, family, man, woman, contract, treatment	52.80%	5.89%	People stories	50.79%	(1) Nursing homes and elderly victims (2) Mental health impact (3) Jail system and cops (4) Family stories
government, country, read, disease, national, announce, area, list, official, include	45.41%	7.30%	Authorities & Politics	29.25%	(1) Health authorities (2) Political authorities (3) Elections
amp, watch, pm, today, fight, post, meet, live, pandemic, speak	50.64%	3.29%			
news, update, late, live, city, follow, resident, outbreak, breaking, local	38.98%	2.82%			
response, pandemic, trump, crisis, american, call, president, plan, president_trump, deal	50.58%	6.53%			
state, test, positive, govt, centre, maharashtra, coronavirusoutbreak, live_update, district, conduct	52.99%	2.72%			
week, start, month, plan, begin, end, set, announce, march, early	51.05%	6.71%	Preventive measures	32.67%	(1) Mobility restrictions (2) Travelling protocols (3) People against restrictions or breaking rules (4) Non-pharmacological measures (5) Festivities and holidays (6) Easing restrictions
quarantine, test_positive, home, day, travel, return, leave, send, stay, passenger	49.47%	8.02%			
people, order, public, urge, issue, social_distance, police, follow, guideline, place	51.59%	3.98%			
lockdown, restriction, rule, extend, ease, measure, level, break, lift, impose	51.03%	4.66%			
outbreak, virus, spread, warn, global, official, country, world, emergency, vanguardnews	37.59%	5.53%	Virus spreading	34.40%	(1) Outbreak and pandemic origins (2) Research and treatment (3) New waves and variants (4) Health workers
people, find, expert, virus, risk, study, infect, variant, claim, spread	48.63%	2.86%			
vaccine, dose, receive, astrazeneca, give, trial, supply, ready, jab, shoot	61.93%	10.50%	Vaccines and vaccination	60.39%	None



Emotion Analysis

Emotion Models

- Plutchik's model postulates that **basic emotions** occur in **opposite pairs**, and **complex emotions** are produced by their **combinations**, naming eight such fundamental emotions.



Ex: Suppose we give the model this tweet:

"I finally passed my exam! I'm so happy!"

The input is simply the text of the tweet. The model processes the words and produces a score for each of four possible emotions:

Emotion	Softmax score
Joy	0.82
Optimism	0.10
Anger	0.03
Sadness	0.05



Because Joy has the highest score (0.82), which is **above the chosen 0.50 threshold**, the final prediction is:

Tweet → Joy 😊

Sentiment Analysis VS. Emotion Detection

Both analyse how people express themselves in text, but they answer different questions.

1. Sentiment analysis: “Is the text **positive or negative**?”

Sentiment analysis is broader and simpler. It determines the **general polarity** or overall attitude expressed in a piece of text.

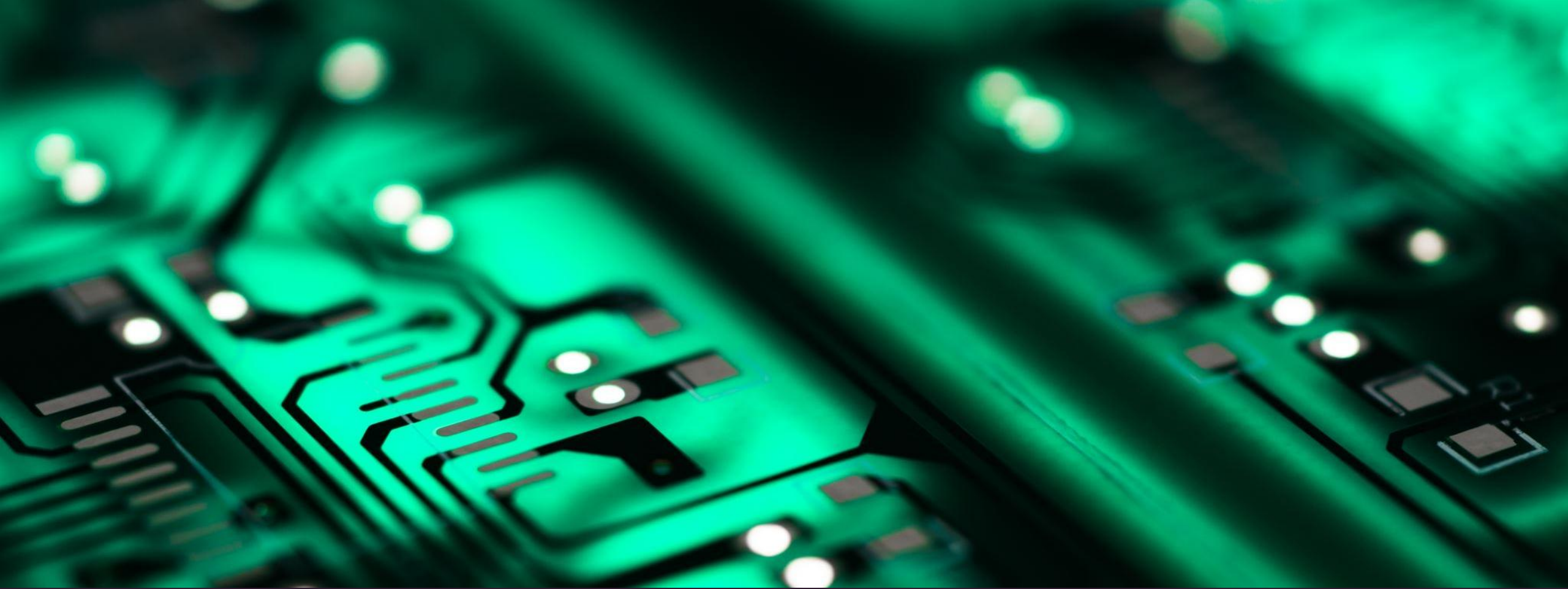
2. Emotion detection: “What specific emotion is being expressed?”

Emotion detection goes **one level deeper**. Instead of simply saying that something is positive or negative, it attempts to identify the **specific emotional state**.

Text	Sentiment Analysis	Emotion Detection
“I got the scholarship!”	Positive	Joy
“I think everything will work out.”	Positive	Optimism
“This makes me furious.”	Negative	Anger
“I really miss my family.”	Negative	Sadness

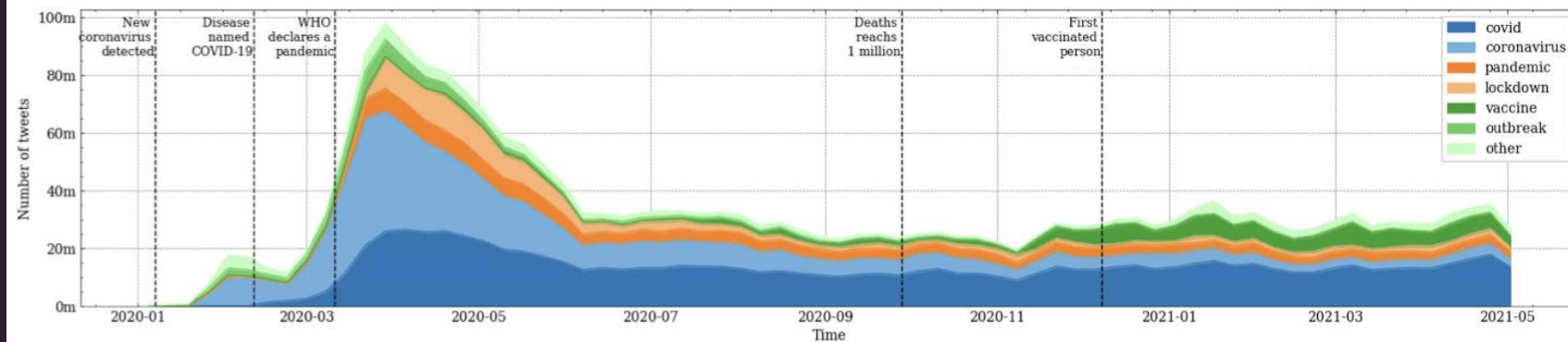
Examples for Emotion Classification

Tweet content	Country	Emotion	Score
"... He doesn't respect the reporters neither the people that are at the frontline fighting for this virus ..."	United States	anger	0.937
"... so many doctors in Europe who have to decide who gets a respirator and who has to die ..."	United States	sadness	0.899
"... I do hope you and the rest of the 1743 family are well ..."	United Kingdom	optimism	0.556
"... I heard this as well via a reputable news source. You're not imagining it. :) ..."	United Kingdom	joy	0.581

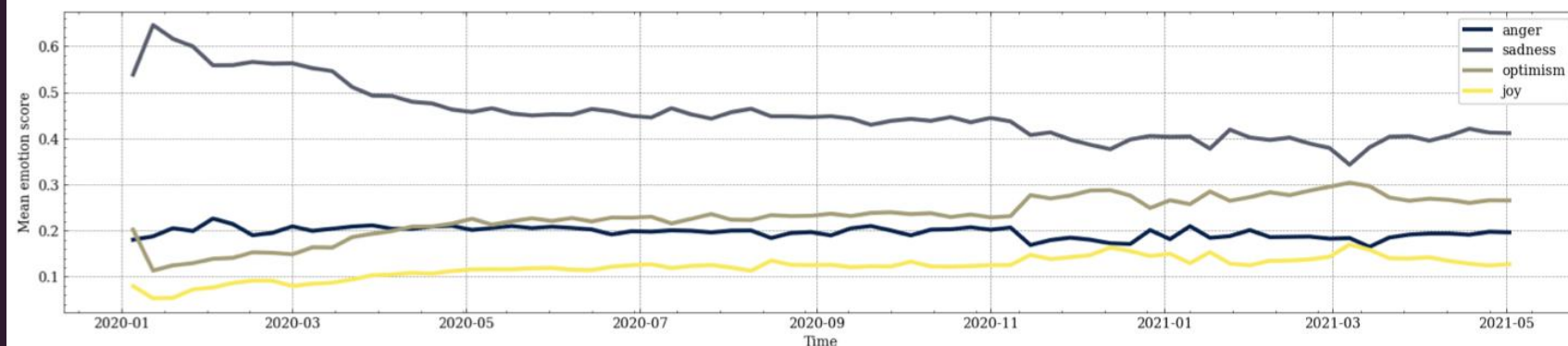


Findings

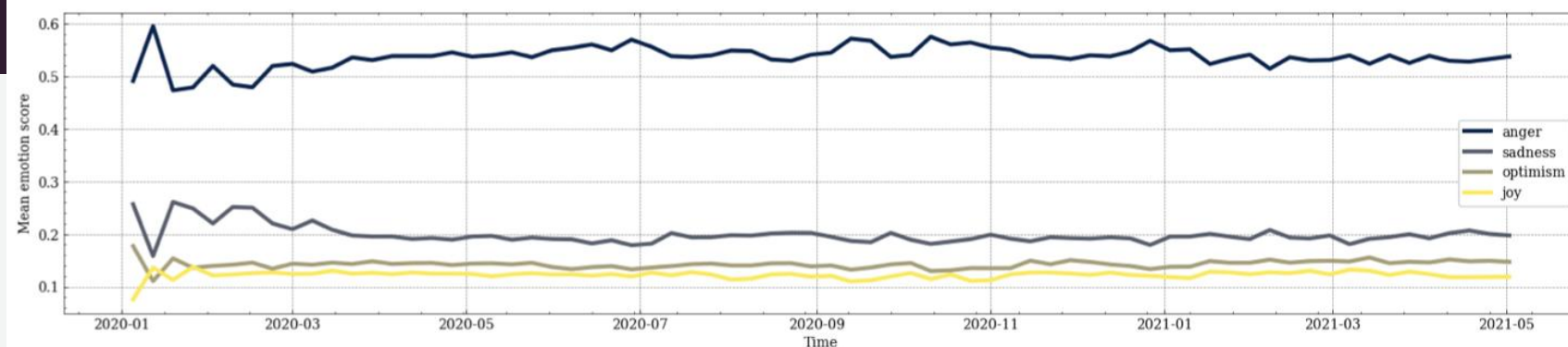
A longitudinal Analysis of the Collected Tweets



(a) The distribution of the COVID-19-related keywords over time.



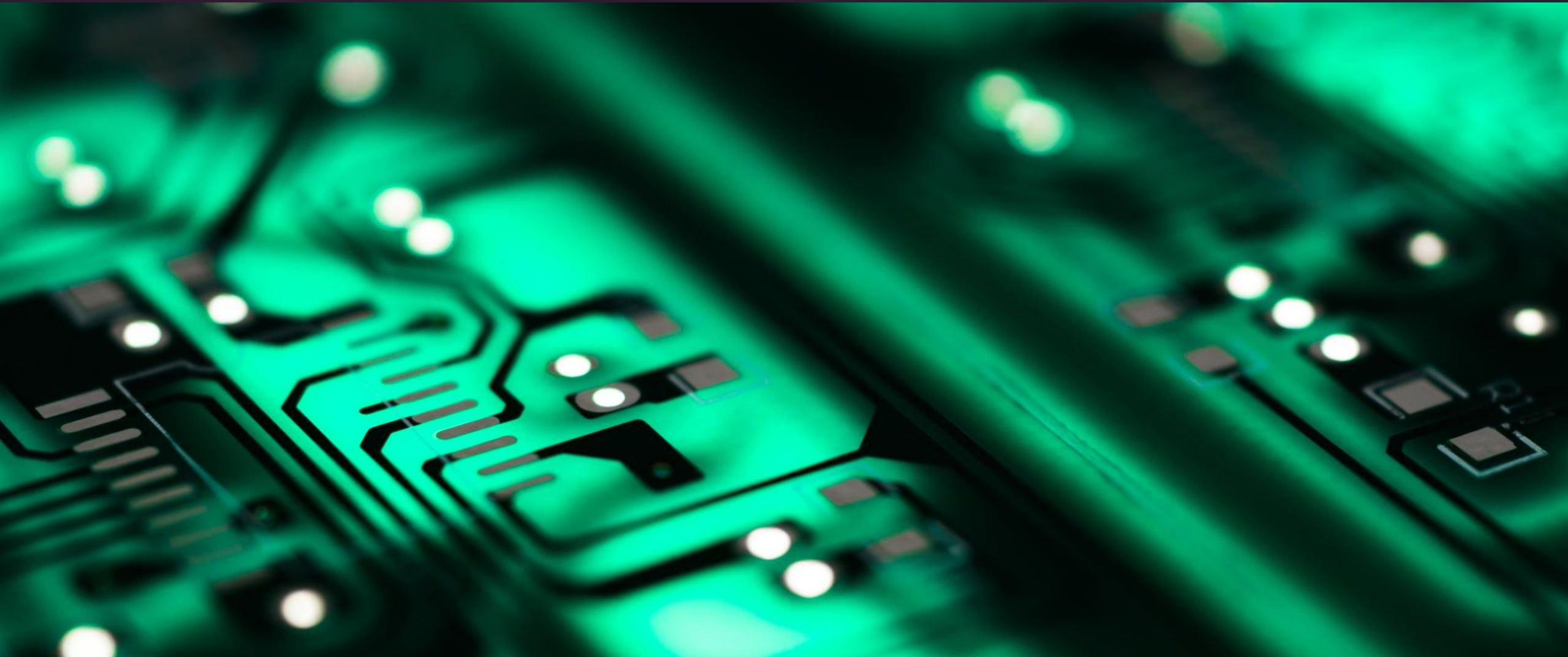
(b) The prevalence of the emotions in the COVID-19 news tweets over time.



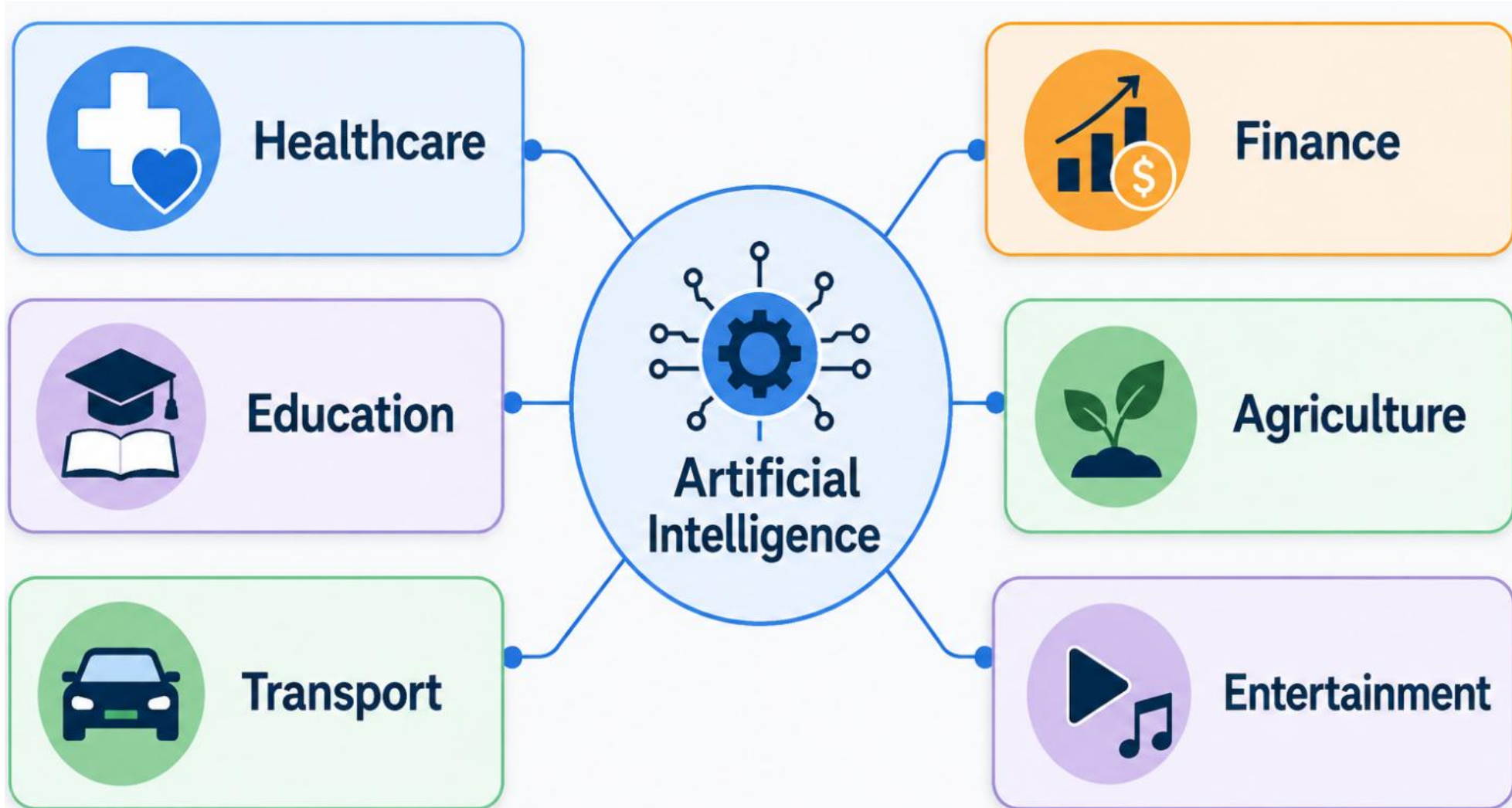
(c) The prevalence of the emotions in the user responses to the COVID-19 news tweets over time.

FIGURE 5. A longitudinal analysis of the collected tweets: (a) shows the number of tweets collected per week, the colors correspond to the keywords used for collecting the tweets; “other” denotes less frequent keywords. Also, (b) shows the weekly mean score of each emotion over the news tweets, and (c) shows the weekly mean score of each emotion over the user responses.

Modern AI

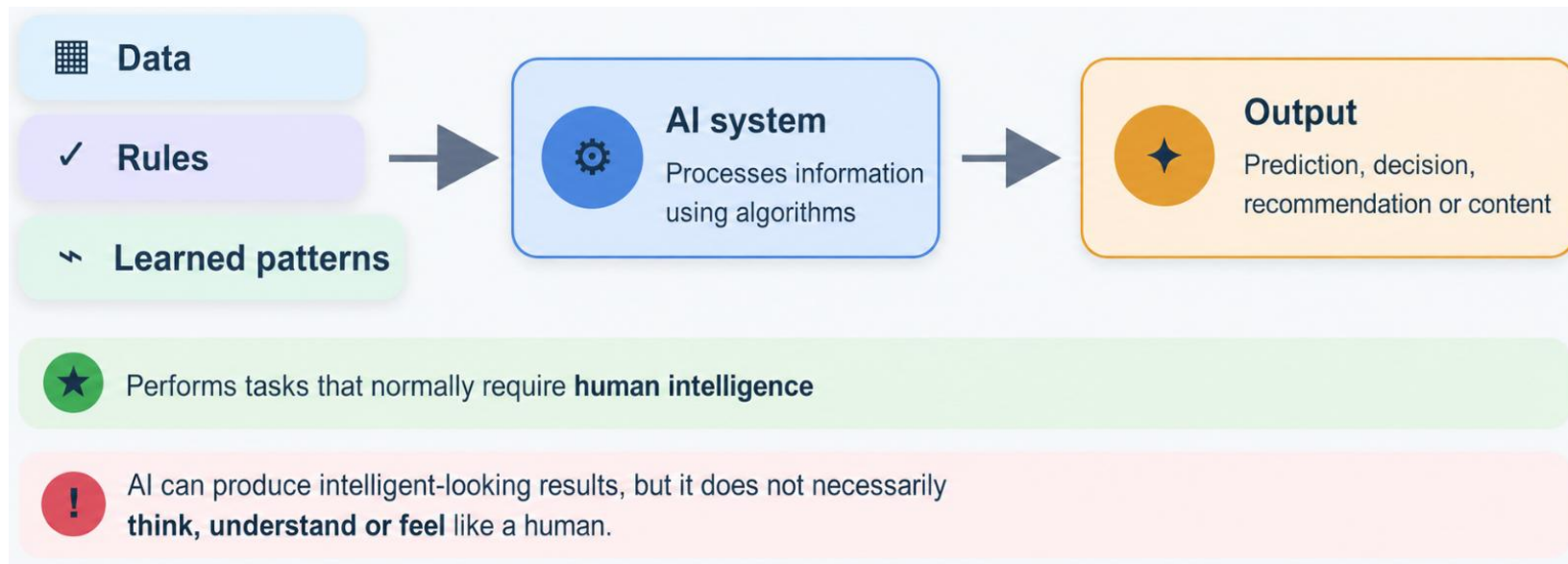


Part 1: AI and its Applications



What Is AI?

Artificial intelligence (AI) is the ability of a computer system to **use data, rules or learned patterns** to produce **outputs** that appear **intelligent**.



AI refers to systems that perform tasks normally **requiring human intelligence**.

What Does “Human Intelligence” Mean in AI?

Human activity	AI-supported equivalent
 Recognising a friend's face	 Facial-recognition system
 Deciding whether an email is suspicious	 Spam-detection system
 Recommending a film to a friend	 Recommendation system
 Translating between languages	 Machine-translation system
 Writing a paragraph	 Generative AI system
 Identifying a disease from an X-ray	 Medical-image classification system

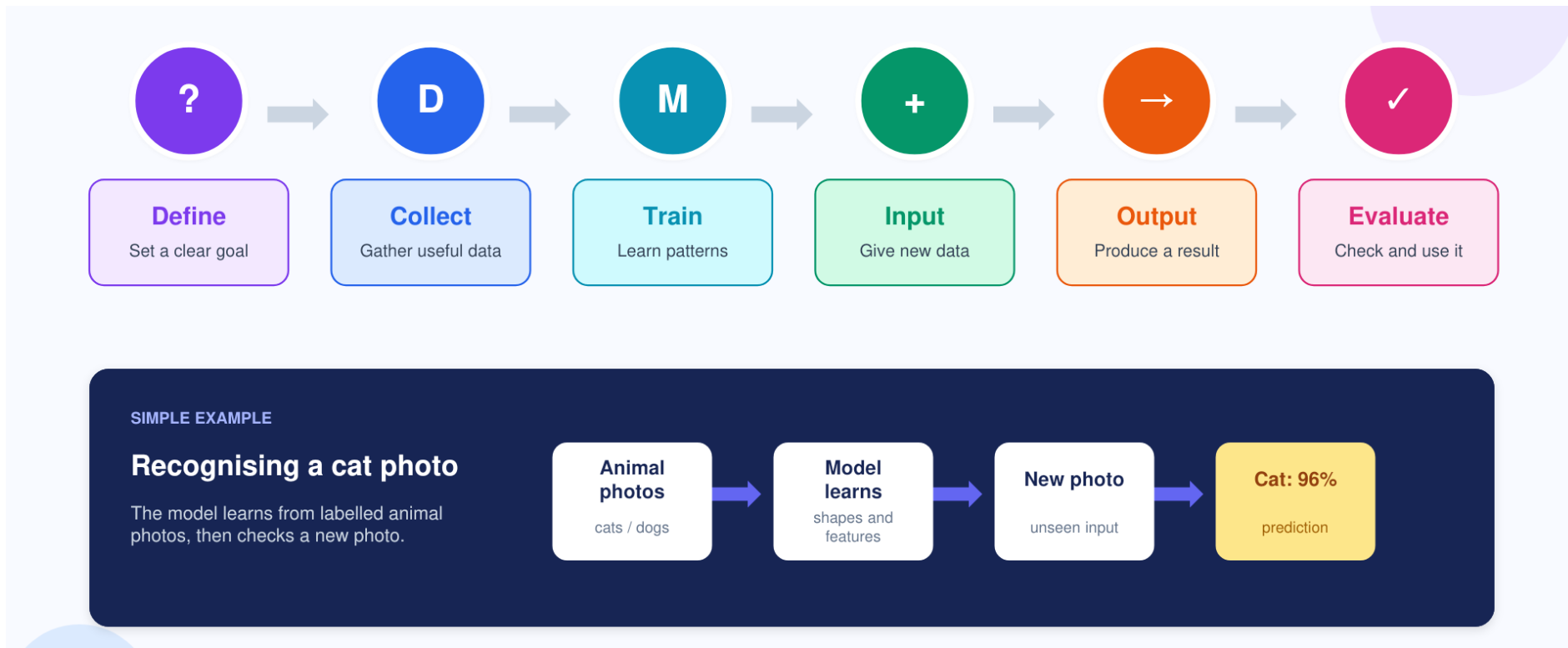
Common AI Capabilities and Applications

AI capability	What it does	Example
 Classification	Assigns an item to a category	Spam-email detection
 Prediction	Estimates a future outcome	Predicting energy demand
 Recommendation	Suggests relevant items	Netflix recommendations
 Computer vision	Analyses images or videos	Facial recognition
 Natural language processing	Analyses human language	Sentiment analysis
 Generative AI	Creates new content	ChatGPT or image generation
 Robotics	Connects AI with physical actions	Warehouse robots

How an AI System Works

✓ An AI system transforms information into a useful output.

A simplified AI pipeline is:



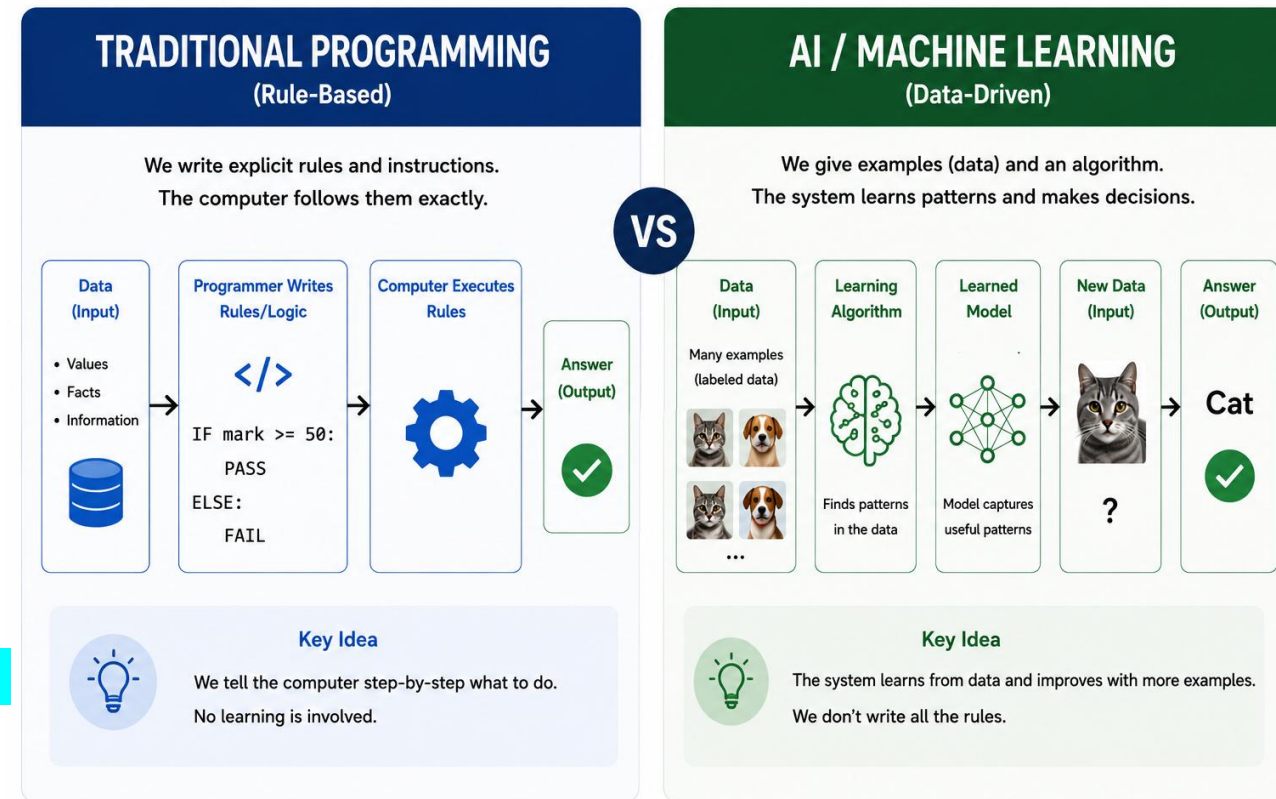
Traditional Programming vs AI

Traditional Programming

- Humans write **explicit rules** and instructions.
- The **computer follows** those rules **exactly**.
- The computer **does not learn** from the data.
- **New situations** often require **new rules** written by the programmer.
- Best for **problems with clear and predictable rules**.

AI

- Humans provide **data** and a **learning algorithm**.
- The **system learns patterns** from examples.
- The learned model can make **predictions on new data**.
- Performance can improve with **more or better-quality data**.
- Useful when **rules are too complex** to write manually.



Automated system vs. AI system

Automation and AI are related, but they are not the same.

A traditional automated system works using rules that have already been defined by a human programmer.

Example:

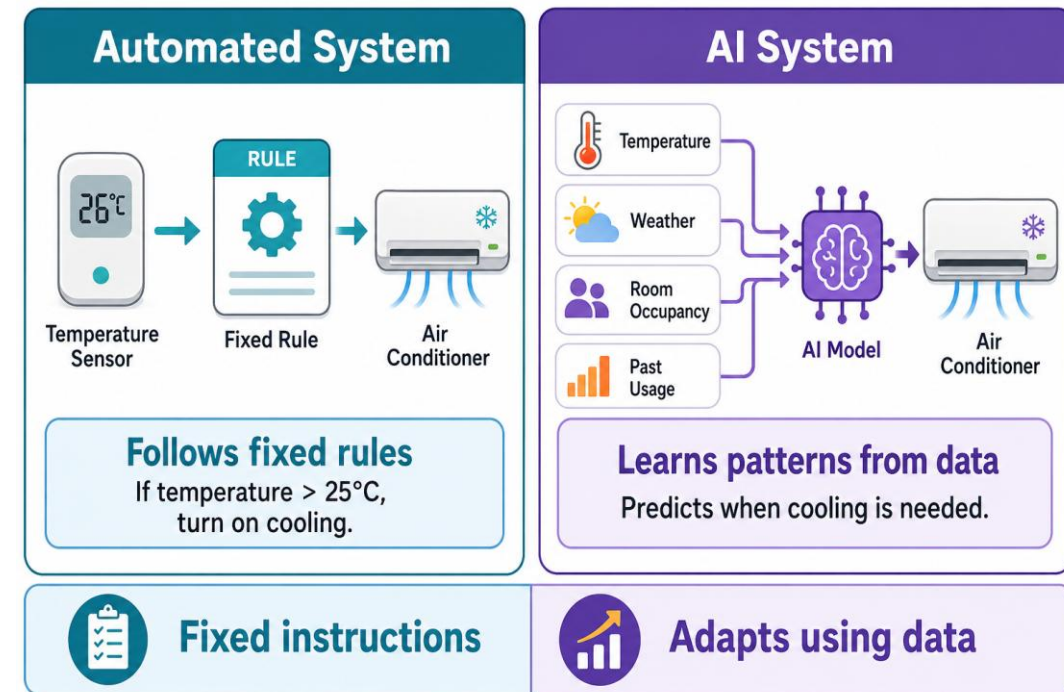
IF temperature > 25°C → Turn ON the air conditioner

ELSE → Keep it OFF

An AI system can make a more sophisticated decision by learning patterns from data rather than relying only on a single predefined rule.

Imagine a smart air-conditioning system. It could use:

1. **Current temperature:** Is the room already getting warm?
2. **Weather forecast:** Will it become much hotter in the next hour?
3. **Room occupancy:** How many people are currently in the room?
4. **User preferences:** What temperature do occupants usually prefer?
5. **Historical behaviour:** When do people normally arrive and leave?



Predictive vs. Generative AI

Predictive AI: Uses existing data to predict outcomes, classify information, or make recommendations.

Generative AI: Uses learned patterns to create new content such as text, images, audio, or code.

	 Predictive AI	 Generative AI
 Purpose	 Predicts a category, value, or future outcome	 Creates new content from learned patterns
 Input	 Past data and a new case	 A prompt or instruction
 Email	 Predicts if an email is spam	 Writes a new email reply
 Image	 Identifies a cat in a photo	 Creates a new cat image from a prompt

Real-world application areas

Healthcare: disease detection and medical-image analysis

Education: personalised learning and automated feedback

Cybersecurity: detecting suspicious activity

Finance: fraud detection and credit assessment

Transportation: navigation and driver-assistance systems

Retail: recommendation systems and demand forecasting

Agriculture: crop monitoring and weed detection

Accessibility: speech recognition, captioning and assistive technologies

Limitations of AI



Limited understanding

AI does not understand as humans do.



Data dependence

Performance depends heavily on data.



Accuracy $\neq$ fairness

High accuracy does not guarantee fairness.



Convincing but incorrect

Generative AI can produce false information.



Lab $\neq$ real world

Laboratory performance may not transfer to real life.



Human judgement matters

People must review important outputs.

Classroom Activity: Intelligent or Just Automated?

You are an **AI investigation team**. Your task is to examine six everyday technologies and decide whether they use AI or whether they simply follow predefined rules.

Technology cards



Google Maps



Calculator



Spotify



Automatic door



ChatGPT



Facial-recognition system

Not every smart-looking system uses AI. Look at how it processes data—not only at what it does

References

- <https://www.sciencedirect.com/science/article/pii/S2468696423000125>
- <https://ieeexplore.ieee.org/abstract/document/9709267>
- <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8652655/>
- <https://www.hindawi.com/journals/hbet/2023/7283166/>
- https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4383670



End of lecture ...